Edge Cases – Multi-Currency & Decimals

1. Decimal Precision
   Top-up of 12.345 → round to 12.35 (two decimal places)
   Payment of 0.001 → reject if less than smallest unit

POST http://localhost:3000/api/wallets/f7c32b8f-65f7-452e-a6bf-79675426d9ef/topup

Send

Body · Headers 1 · JSON · Prettify

```
{
  "amount": 12.345,
  "reference_id": "tx-unique-12345"

}
```

Response · Headers 19 · JSON · 200 OK 89ms 158B

```
{
  "message": "Topup berhasil !",
  "data": {
    "id": "f7c32b8f-65f7-452e-a6bf-79675426d9ef",
    "owner_id": "user-003",
    "currency": "USD",
    "balance": "12.35",
    "status": "ACTIVE"
  }
}
```

POST http://localhost:3000/api/wallets/f7c32b8f-65f7-452e-a6bf-79675426d9ef/payment

Send

Params · Body · Headers 1 · JSON · Prettify

```
{
  "amount": 0.001,
  "reference_id": "pay-order-0001"
}
```

Response · JSON · 400 Bad Request 79ms 74B

```
{
  "error": "Presisi jumlah tidak valid (maksimal 2 angka di belakang koma)"
}
```

2. Large Balances
   System must safely store 1,000,000,000.00 or higher

POST http://localhost:3000/api/wallets/f7c32b8f-65f7-452e-a6bf-79675426d9ef/topup

Send

Params · Body · Headers 1 · JSON · Prettify

```
{
  "amount": 1000000000.00,
  "reference_id": "tx-unique-1m"

}
```

Response · Headers 19 · JSON · 200 OK 36ms 166B

```
{
  "message": "Topup berhasil !",
  "data": {
    "id": "f7c32b8f-65f7-452e-a6bf-79675426d9ef",
    "owner_id": "user-003",
    "currency": "USD",
    "balance": "1000000012.35",
    "status": "ACTIVE"
  }
}
```

3. Currency Mismatch
   Transfers between wallets of different currencies are rejected

POST http://localhost:3000/api/wallets/transfer

Send

Params | Body | Headers 1 | JSON | Prettify

```
{
  "sender_id": "8b38fd82-a808-432e-950a-fd2f43747872",
  "receiver_id": "f36db4fd-a7f3-4d6a-966c-235f2773f49d",
  "amount": 40,
  "reference_id": "trf-unique-999"
}
```

Response | JSON | 400 Bad Request 64ms 59B

```
{
  "error": "Transfer antar mata uang berbeda belum didukung"
}
```

4. Multiple Wallets Per User

   Users can hold multiple currencies, but one wallet per currency

POST http://localhost:3000/api/wallets

Send

Params | Body | Headers 1 | JSON | Prettify

```
{
  "owner_id": "user-004",

  "currency": "USD"
}
```

Response | JSON | 201 Created 97ms 179B

```
{
  "status": "success",
  "message": "Wallet berhasil dibuat",
  "data": {
    "id": "8b38fd82-a808-432e-950a-fd2f43747872",
    "owner_id": "user-004",
    "currency": "EUR",
    "balance": "0",
    "status": "ACTIVE"
  }
}
```

POST http://localhost:3000/api/wallets

Send

Params | Body | Headers 1 | JSON | Prettify

```
{
  "owner_id": "user-004",

  "currency": "USD"
}
```

Response | JSON | 201 Created 141ms 179B

```
{
  "status": "success",
  "message": "Wallet berhasil dibuat",
  "data": {
    "id": "649d773f-deb3-4193-8914-9c3b938df7de",
    "owner_id": "user-004",
    "currency": "USD",
    "balance": "0",
    "status": "ACTIVE"
  }
}
```

POST http://localhost:3000/api/wallets

Send

Params | Body | Headers 1 | JSON | Prettify

```
{
  "owner_id": "user-004",

  "currency": "USD"
}
```

Response | JSON | 409 Conflict 129ms 64B

```
{
  "error": "User sudah memiliki wallet aktif untuk mata uang USD"
}
```

5. Zero or Negative Amounts

   Reject operations with 0.00 or negative amounts

POST http://localhost:3000/api/wallets/8b38fd82-a808-432e-950a-fd2f43747872/payment

Send

Params Body Headers JSON Prettify

```
{
  "amount": 0.00,

  "reference_id": "pay-order-0001"
}
```

Response JSON 400 Bad Request 23ms 282B

```
{
  "error": {
    "name": "ZodError",
    "message": "[\n  {\n    \"origin\": \"number\",\n    \"code\": \"too_small\",\n    \"minimum\": 0,\n    \"inclusive\": false,\n    \"path\": [\n      \"body\",\n      \"amount\"\n    ],\n    \"message\": \"Jumlah pembayaran harus lebih dari 0\"\n  }\n]"
  }
}
```

6. Duplicate Requests
   Safely ignore double top-ups or payments

POST http://localhost:3000/api/wallets/8b38fd82-a808-432e-950a-fd2f43747872/payment

Send

Params Body Headers JSON Prettify

```
{
  "amount": 0.11,
  "reference_id": "pay-order-0001"
}
```

Response JSON 409 Conflict 32ms 47B

```
{
  "error": "Reference ID sudah pernah digunakan"
}
```

7. Concurrent Spending
   Multiple payments at same time must not allow negative balance

POST http://localhost:3000/api/wallets/e33774b2-5e58-47e3-990b-89575c9346ea/payment

Send

Params Body Headers JSON Prettify

```
{
  "amount": 1600.00,
  "reference_id": "pay-order-56"

}
```

Response JSON 400 Bad Request 42ms 29B

```
{
  "error": "Saldo tidak cukup"
}
```

8. Partial Failure During Transfer
   Debit and credit must be atomic

```
await tx.wallet.update({
  where: { id: senderId },
  data: { balance: Decimal.sub(sender.balance, transferAmount) },
});
await tx.wallet.update({
  where: { id: receiverId },
  data: { balance: Decimal.add(receiver.balance, transferAmount) },
});

await tx.ledger.createMany({
  data: [
    {
      wallet_id: senderId,
      transaction_type: "TRANSFER_OUT",
      amount: transferAmount.negated(),
      currency: sender.currency,
      reference_id: `${referenceId}-OUT`
    },
    {
      wallet_id: receiverId,
      transaction_type: "TRANSFER_IN",
      amount: transferAmount,
      currency: receiver.currency,
      reference_id: `${referenceId}-IN`
    }
  ],
```

.

9. Ledger vs Balance Mismatch
   Wallet balance must always match sum of ledger entries

Response Headers [19] Timeline Tests


```
1  {
2    "data": [
3      {
4        "id": "3864687d-20f2-4651-af50-965205ed5c12",
5        "wallet_id": "8b38fd82-a808-432e-950a-fd2f43747872",
6        "transaction_type": "TRANSFER_IN",
7        "amount": "50",
8        "currency": "EUR",
9        "reference_id": "trf-unique-SAGFS-IN",
10       "created_at": "2026-04-29T09:39:17.026Z"
11     },
12     {
13       "id": "804751da-ad18-4ebd-b8c7-0c0132c6470f",
14       "wallet_id": "8b38fd82-a808-432e-950a-fd2f43747872",
15       "transaction_type": "TRANSFER_OUT",
16       "amount": "-100",
17       "currency": "EUR",
18       "reference_id": "trf-unique-54GE-OUT",
19       "created_at": "2026-04-29T09:38:21.432Z"
20     },
21     {
22       "id": "385a994e-5d21-4656-9398-dd332dddca06",
23       "wallet_id": "8b38fd82-a808-432e-950a-fd2f43747872",
24       "transaction_type": "PAYMENT",
25       "amount": "-43.21",
26       "currency": "EUR",
27       "reference_id": "pay-order-214j",
28       "created_at": "2026-04-29T09:37:13.716Z"
29     },
30     {
31       "id": "9805fae7-c24b-40e2-897b-97ffc3bd7d2a",
32       "wallet_id": "8b38fd82-a808-432e-950a-fd2f43747872",
33       "transaction_type": "TOPUP",
34       "amount": "543.21",
35       "currency": "EUR",
36       "reference_id": "tx-unique-asd",
37       "created_at": "2026-04-29T09:05:58.355Z"
38     }
39   ]
40 }
```

Response Headers 19 Timeline Tests


```
{
  "data": {
    "id": "8b38fd82-a808-432e-950a-fd2f43747872",
    "owner_id": "user-004",
    "currency": "EUR",
    "balance": "450",
    "status": "ACTIVE",
    "_count": {
      "ledgers": 4
    }
  }
}
```


10. Suspended Wallet Operations
    Suspended wallets cannot be topped-up, paid from, or transferred

patch http://localhost:3000/api/wallets/8b38fd82-a808-432e-950a-fd2f43747872/status Send

Params Body * Headers 1 JSON Prettify


```
{
  "status": "SUSPENDED"

}
//ENUM = ACTIVE | SUSPENDED
```


Response Headers 19 JSON 200 OK 64ms 174B


```
{
  "message": "Wallet berhasil di-suspended",
  "data": {
    "id": "8b38fd82-a808-432e-950a-fd2f43747872",
    "owner_id": "user-004",
    "currency": "EUR",
    "balance": "543.21",
    "status": "SUSPENDED"
  }
}
```


get http://localhost:3000/api/wallets/8b38fd82-a808-432e-950a-fd2f43747872 Send

Params Body Headers Auth * Vars Script

Query

Name	Value
Name	Value

Bulk Edit

Path

Response Headers 19 JSON 200 OK 13ms 156B


```
{
  "data": {
    "id": "8b38fd82-a808-432e-950a-fd2f43747872",
    "owner_id": "user-004",
    "currency": "EUR",
    "balance": "543.21",
    "status": "SUSPENDED",
    "_count": {
      "ledgers": 1
    }
  }
}
```


11. Out-of-Order Requests
    Operations must remain consistent regardless of request order

```
const debitAmount = transferAmount.negated();
const creditAmount = transferAmount;

await tx.wallet.update({
  where: { id: senderId },
  data: {
    balance: {
      increment: debitAmount
    }
  },
});
await tx.wallet.update({
  where: { id: receiverId },
  data: {
    balance: {
      increment: creditAmount
    }
  },        You, 10 minutes ago • Uncommitted changes
});
```

```
const ledgerAmount = payAmount.negated();

const updatedWallet = await tx.wallet.update({
  where: { id: walletId },
  data: {
    balance: {
      increment: ledgerAmount
    },
  },
});
```

12. Read-After-Write Consistency
    Queries must reflect the latest committed balance

13. System Restart / Crash Recovery
    Partial transactions must not leave wallets in inconsistent state